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Hazard Pointer Batches

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Introduction

The Varna 2023 plenary voted in favor of including hazard pointers in the C++26 standard library ([2023-06 LWG Motion 7] P2530R3 Hazard Pointers for C++26).

P3135R1 was presented to SG1 in Tokyo 2024, reviewing potential extensions of the P2530R3 C++26 interface, and proposing two of those for inclusion in the standard library. The proposal for extending the P2530R3 C++26 interface to support synchronous reclamation was voted on by the Concurrency Study Group (SG1) in Tokyo 2024 as follows:

```
we want to continue work on hazard pointer batches for C++26, except as free-functions
that take and return an existing type of collection (e.g. std::array, or a range,
or...)
SF F N A SA
 3 7 2 0 0
Unanimous consent
```

This paper is a follow up on P3135R1, focusing on extending the P2530R3 C++26 hazard pointer interface to support creation and destruction of batches of nonempty hazard pointers, revised to take into account the feedback from SG1.

Background: P2530R3 C++26 Hazard Pointers

Hazard pointer interface from P2530R3:

```
template <class T, class D = default_delete<T>>
class hazard_pointer_obj_base {
public:
    void retire(D d = D()) noexcept;
protected:
    hazard_pointer_obj_base() = default;
    hazard_pointer_obj_base(const hazard_pointer_obj_base&) = default;
    hazard_pointer_obj_base(hazard_pointer_obj_base&&) = default;
    hazard_pointer_obj_base& operator=(const hazard_pointer_obj_base&) = default;
    hazard_pointer_obj_base& operator=(hazard_pointer_obj_base&&) = default;
    ~hazard_pointer_obj_base() = default;
private:
    D deleter ; // exposition only
};

class hazard_pointer {
public:
    hazard_pointer() noexcept; // Constructs an empty hazard_pointer
    hazard_pointer(hazard_pointer&&) noexcept;
```

```

hazard_pointer& operator=(hazard_pointer&&) noexcept;
~hazard_pointer();
[[nodiscard]] bool empty() const noexcept;
template <class T> T* protect(const atomic<T*>& src) noexcept;
template <class T> bool try_protect(T*& ptr, const atomic<T*>& src) noexcept;
template <class T> void reset_protection(const T* ptr) noexcept;
void reset_protection(nullptr_t = nullptr) noexcept;
void swap(hazard_pointer&) noexcept;
};

```

```

hazard_pointer make_hazard_pointer(); // Constructs a nonempty hazard_pointer
void swap(hazard_pointer&, hazard_pointer&) noexcept;

```

Brief notes (See P2530R3 for details):

- A `hazard_pointer` object is either *empty* or *nonempty*. It is nonempty if and only if it owns a hazard pointer. Only nonempty `hazard_pointer` objects can be used to protect protectable objects.
- The default constructor of `hazard_pointer` constructs an empty object, whereas the free function `make_hazard_pointer` constructs a nonempty object.

Motivation

The construction and destruction of multiple nonempty `hazard_pointer` objects in one batch has lower latency than their construction and destruction separately, e.g., 2 ns vs 6 ns for the construction/destruction of 3 nonempty hazard pointers.

Implementation and Use Experience

Batches of hazard pointers have been part of the Folly open-source library (under the name `hazptr_array` as a distinct class) and in heavy use in production since 2017.

Batches of Hazard Pointers

The construction and destruction of a nonempty `hazard_pointer` object typically involves access to thread-local storage and has low but non negligible latency (low single digit nanoseconds). The P2530R3 C++26 hazard pointer interface supports only the construction and destruction of nonempty hazard pointers individually.

This paper proposes adding support for the construction and destruction of multiple hazard pointers in one batch to reduce the latency of such operations.

Possible Interface

```
/* Takes a span of empty hazard_pointer objects and makes them nonempty. */
void make_hazard_pointer_batch(std::span<hazard_pointer> span);

/* Takes a span of nonempty hazard_pointer objects and makes them empty. */
void empty_hazard_pointer_batch(std::span<hazard_pointer> span) noexcept;
```

Usage Example

The following table shows two functionally-equivalent code snippets using the P2530R3 C++26 hazard pointer interface and using hazard pointer batches.

P2530R3 C++26	Hazard Pointer Batches
<pre>{ hazard_pointer hp[3]; /* Three hazard pointers are made nonempty separately. */ hp[0] = make_hazard_pointer(); hp[1] = make_hazard_pointer(); hp[2] = make_hazard_pointer(); assert(!hp[0].empty()); assert(!hp[1].empty()); assert(!hp[2].empty()); // src is atomic<T*> T* ptr = hp[0].protect(src); /* etc */ } /* Three nonempty hazard pointers are destroyed separately. */</pre>	<pre>{ hazard_pointer hp[3]; /* Three hazard pointers are made nonempty together. */ make_hazard_pointer_batch(hp); assert(!hp[0].empty()); assert(!hp[1].empty()); assert(!hp[2].empty()); /* src is atomic<T*> */ T* ptr = hp[0].protect(src); /* etc */ empty_hazard_pointer_batch(hp); /* Three nonempty hazard pointers are emptied together. */ } /* The three emptied hazard pointers are destroyed separately. */</pre>

References

- [P2530R3](#): Hazard Pointers for C++26 (2023-03-02).

- [P3135R1](#): Hazard Pointer Extensions (2024-04-12).
- [Folly](#): Facebook Open-source Library.